

RAIPUR, India

WMO: 428740

Lat: 21.233N

Lon: 81.650E

Elev: 978

StdP: 14.18

Time Zone: 5.50 (IND)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	54.8	56.9	35.2	31.2	82.2	38.3	35.3	80.7	7.5	72.4	6.5	73.7	0.7	50	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.6	109.8	72.5	107.8	72.8	105.6	72.5	80.9	89.7	80.1	88.5	79.6	87.8	4.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	156.3	84.7	78.3	152.3	83.7	77.8	149.7	83.2	45.7	90.3	44.9	88.6	44.3	88.1	89.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	9.8	8.0	DB	50.4	112.9	2.0	2.1	49.0	114.4	47.8	115.7	46.7	116.9	45.2	118.4	(4)
(5)			WB	46.3	83.1	2.1	1.4	44.8	84.1	43.5	84.9	42.4	85.6	40.9	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	70.5	76.1	83.6	90.9	95.2	89.3	82.1	81.3	82.2	80.3	75.3	70.5		
	DBStd	8.35	3.65	4.42	4.54	4.19	3.68	6.59	3.31	2.56	2.14	2.73	3.31	3.32		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	6	3	1	0	0	0	0	0	0	0	0	0	2		
	CDD50	11481	637	732	1041	1228	1402	1180	995	969	966	938	758	635		
	CDD65	6011	175	313	576	778	937	730	530	504	516	473	308	171		
	CDH74	74270	1358	3058	7367	12129	15603	10583	5591	4997	5453	4519	2403	1209		
	CDH80	39499	340	1369	4168	8048	11178	6543	1906	1405	1849	1690	775	228		
(14) Wind	WSAvg	2.6	1.3	2.0	2.2	3.1	3.5	4.5	4.5	3.7	2.7	1.7	1.2	1.0		
(15) Precipitation	PrecAvg	48.7	0.4	0.5	0.7	0.5	0.8	6.9	14.5	13.1	8.6	2.2	0.2	0.3		
	PrecMax	64.8	4.3	2.8	1.9	4.4	4.4	24.9	27.5	20.5	17.5	6.9	2.4	2.6		
	PrecMin	31.3	0.0	0.0	0.1	0.0	0.0	0.9	4.3	6.4	1.6	0.1	0.0	0.0		
	PrecStd	9.1	0.9	0.7	0.4	0.9	1.0	4.8	5.6	3.8	3.8	2.0	0.5	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	96.1	103.4	109.4	113.0	111.4	97.0	91.7	92.8	92.8	90.1	87.1		
		MCWB	65.7	66.3	68.7	70.8	71.9	73.4	79.1	78.9	77.4	73.9	71.0	67.1		
	2%	DB	85.8	92.8	100.7	107.4	111.1	108.6	92.8	89.7	90.8	90.9	88.0	84.5		
		MCWB	65.2	67.0	69.1	71.1	71.8	74.6	78.8	78.7	77.8	74.0	70.1	65.9		
	5%	DB	83.7	90.5	98.7	105.7	109.0	105.5	90.3	88.2	89.4	89.4	86.4	82.5		
		MCWB	64.4	66.4	68.8	70.4	72.5	75.8	78.9	78.7	78.2	74.2	69.3	64.9		
	10%	DB	81.5	88.1	96.3	103.6	106.9	101.6	88.2	86.5	87.9	87.8	84.7	80.6		
		MCWB	63.3	65.9	68.5	70.6	72.8	76.3	78.7	78.4	78.2	73.7	68.5	64.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.7	74.2	76.5	78.8	80.4	82.3	82.5	81.0	81.3	79.6	76.3	71.5		
		MCDB	79.1	85.9	95.7	99.7	101.2	95.1	90.0	87.3	87.6	86.6	85.3	81.1		
	2%	WB	68.7	71.3	74.1	76.9	78.8	80.8	80.7	80.3	80.3	78.7	74.7	69.7		
		MCDB	78.1	85.6	92.5	97.3	99.3	92.9	88.1	86.2	86.8	85.8	82.3	78.9		
	5%	WB	66.9	69.4	72.4	75.4	77.6	79.8	79.8	79.6	79.8	77.9	73.3	68.2		
		MCDB	78.6	83.5	90.2	96.2	97.9	91.7	86.8	85.4	86.1	84.7	80.6	76.8		
	10%	WB	65.5	68.1	71.1	74.0	76.5	79.0	79.1	79.0	79.1	77.1	71.8	66.6		
		MCDB	78.1	82.2	88.8	94.7	96.8	90.8	85.8	84.4	85.2	83.5	80.0	76.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	22.4	23.0	23.8	22.6	16.2	9.6	8.6	11.0	15.5	20.6	22.1		
		MCDBR	24.6	25.0	25.5	25.5	24.3	21.3	13.6	11.4	13.4	17.4	21.4	23.9		
	5% WB	MCWBR	9.4	7.8	7.6	6.9	6.1	4.7	4.2	3.7	4.5	5.2	7.3	9.3		
		MCWBR	20.0	20.4	22.0	23.0	21.8	16.4	11.6	10.2	11.6	12.8	15.8	18.8		
(40) Clear-Sky Solar Irradiance	taub		0.467	0.477	0.536	0.557	0.602	0.658	0.597	0.557	0.550	0.530	0.516	0.475		
	taud		2.016	2.010	1.888	1.887	1.813	1.737	1.880	2.004	2.003	1.994	1.942	2.010		
	Ebn at Noon		251	258	248	245	232	217	231	242	241	240	234	243		
	Edh at Noon		51	54	64	65	69	74	64	57	56	55	55	50		
(44) All-Sky Solar Radiation	RadAvg		1435	1698	1960	2167	2173	1658	1260	1242	1485	1599	1468	1366		
	RadStd		76	72	94	79	50	167	150	118	141	127	81	69		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page